



IT582: Foundations of Machine Learning

Lab-5: Cross-Validation and Bootstrap Methods

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Submission Instructions

Upload **only one Jupyter Notebook (.ipynb)** file to Moodle.

The notebook should contain all code, outputs, plots, observations and conclusions.

Name the notebook as:

[studentid]_lab5.ipynb

Implement the resampling procedures from scratch. Do not use built-in implementations of KFold, LeaveOneOut, or bootstrap utilities.

Task 1: k -Fold Cross-Validation

Generate

$$n = 120, \quad x \in [-2, 2]$$

uniformly, with

$$y = 1 + 0.8x - 1.5x^2 + 0.5x^3 + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.3^2).$$

Keep 20% of the data as an independent test set and use the remaining data for cross-validation.

Consider polynomial regression models with

$$p \in \{1, 2, 3, 5, 9\}.$$

Implement k -fold cross-validation for

$$k = 5 \quad \text{and} \quad k = 10.$$

For each degree and each value of k :

- Train using $k - 1$ folds and evaluate on the remaining fold.
- Repeat until each fold has been used once for validation.

- Record the MSE for each fold and compute the mean CV MSE.

Use

$$CV = \frac{1}{k} \sum_{r=1}^k MSE_r.$$

Plot polynomial degree versus mean CV MSE for both $k = 5$ and $k = 10$

Select the degree with the lowest CV MSE, fit the model using all non-test data, and plot train and test MSE.

Write observations on model capacity and the difference between 5-fold and 10-fold cross-validation.

Task 2: Regression with k -Fold Cross-Validation on a Real Dataset

Use the **Diabetes Regression Dataset** available through

```
sklearn.datasets.load_diabetes()
```

Use all available input features and the provided continuous target.
Implement

$$k = 5$$

fold cross-validation from scratch.

For each fold:

- Use one fold as the validation set and the remaining folds as the training set.
- Standardize the input features using the mean and standard deviation calculated from the training fold only.
- Apply the same transformation to the corresponding validation fold.
- Fit a Linear Regression model using the training fold.
- Calculate the validation MSE.

Report the MSE obtained for all five folds and calculate

$$CV = \frac{1}{5} \sum_{r=1}^5 MSE_r.$$

Also report the standard deviation of the five validation MSE values.

Repeat the experiment using

$$k = 10.$$

Compare the 5-fold and 10-fold results and write observations on:

- Mean validation error.
- Variation in error across folds.
- Amount of training data available in each fold.
- Why cross-validation gives a more informative performance estimate than relying on a single train–test split.

Task 3: Leave-One-Out Cross-Validation

Consider

$$x^{(i)} = A + w^{(i)},$$

where

$$A = 5, \quad w^{(i)} \sim \mathcal{N}(0, 1), \quad n = 12.$$

Estimate A using the complete dataset:

$$\hat{A} = \frac{1}{n} \sum_{i=1}^n x^{(i)}.$$

Now implement **Leave-One-Out Cross-Validation (LOOCV)**.

For each observation i , leave it out and estimate

$$\hat{A}_{-i} = \frac{1}{n-1} \sum_{j \neq i} x^{(j)}.$$

Use $\hat{A}_{-i}$ to predict the left-out observation and compute

$$e_i^2 = \left(x^{(i)} - \hat{A}_{-i} \right)^2.$$

Calculate

$$CV_{\text{LOO}} = \frac{1}{n} \sum_{i=1}^n e_i^2.$$

Report all $\hat{A}_{-i}$ values and corresponding prediction errors.

Compare the full-data estimate with the leave-one-out estimates and write observations on:

- Why LOOCV has n folds.
- Why the leave-one-out estimates are similar.
- Computational cost compared with ordinary k -fold CV.

Task 4: Bootstrap Method

Consider the paired observations:

Observation	1	2	3	4	5	6	7	8
Before	18	21	16	24	20	19	23	17
After	15	18	15	20	18	17	19	16

Calculate the paired differences

$$d_i = \text{Before}_i - \text{After}_i$$

and the sample mean

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i.$$

Implement bootstrap sampling using

$$B = 1000$$

replications.

For each replication, draw n observations from the difference vector **d with replacement** and compute the bootstrap mean S^{*b} .

Calculate

$$\bar{S}^* = \frac{1}{B} \sum_{b=1}^B S^{*b}$$

and

$$\widehat{\text{Var}}[S] = \frac{1}{B-1} \sum_{b=1}^B (S^{*b} - \bar{S}^*)^2.$$

Also report the bootstrap standard deviation

Plot a histogram of the bootstrap estimates and mark the original sample mean.

Write observations on:

- What the spread of the bootstrap estimates represents.
- How bootstrap differs from k -fold CV and LOOCV.